# Dynamic Pricing Optimization for EV Charging Stations

**Business Scenario**

An EV charging company in Lagos wants to:

- Predict hourly charging demand
- Move from static 25% markup pricing
- Implement dynamic pricing
- Maximize profit without driving away customers

**Goal**

Build a system that **dynamically adjusts EV charging prices** based on predicted demand to maximize profit while maintaining user retention.

**TOOL ARCHITECTURE**

| Tools | What tools will be used for |
|---|---|
| Excel | Financial modeling & sensitivity testing |
| PostgreSQL | Store structured charging data |
| Python | Data simulation, cleaning, ML model |
| Machine learning (Linear regression) | Demand forecasting |
| PowerBi | Executive pricing dashboard |

**SYSTEM PIPELINE**

**Python → PostgreSQL → Python ML → Export Results → Excel → Power BI**

**HOW EACH TOOL WILL BE USED**

**1. Python (Data + ML Layer)**

- Simulate EV charging dataset
- Perform EDA
- Train Linear Regression model
- Derive demand function
- Compute dynamic price mathematically
- Export dynamic pricing dataset

**2. PostgreSQL (Data Engineering Layer)**

- **Create tables:**

    - charging_sessions
    - static_pricing
    - dynamic_pricing
- **Write SQL queries:**
    - Aggregate hourly demand
    - Compare static vs dynamic profit
    - Create views for dashboard use

**3. Excel (Financial Modeling Layer)**

- Import dynamic pricing results
- Create:
    - Profit comparison model
    - Sensitivity analysis table
    - Revenue growth scenario model

**4. Power BI (Executive layer)**

Build a dashboard with:

- Hourly demand prediction
- Static vs dynamic profit comparison
- Profit increase %
- Peak hour identification

# FULL MATHEMATICAL DERIVATION

**Demand Function Assumption**

From regression we estimate:

$$D = a - bP$$

Where:

- $D$= demand (kWh)

- $P$= price per kWh
- $a$= baseline demand
- $b$= price sensitivity (elasticity coefficient)

Since demand decreases when price increases:

$$b > 0$$

**Profit Function**

Let:

- $C$= wholesale electricity cost
- $P$= selling price

Profit is:

$$\prod(P) = (P - C) * D$$

Substitute demand function:

$$\prod(P) = (P - C)(a - bP)$$

Expanding the function:

$$\prod(P) = P(a - bP) - C(a - bP)$$

$$\prod(P) = aP - bP^2 - aC + bCP$$

Rearranging:

$$\prod(P) = -bP^2 + (a + bC)P - aC$$

Take First Derivative:

$$\frac{d\prod}{dP} = -2bP + (a + bC)$$

Set equal to zero:

$$-2bP + (a + bC) = 0$$

Solve for Optimal Price:

$$2bP = (a + bC)$$

$$P = \frac{(a + bC)}{2b}$$

Test if the function is concave:

Second derivative:

$$\frac{d^2\prod}{dP^2} = -2b$$

Since:

$$b > 0$$

Then:

$$-2b < 0$$

Therefore, the function is concave, that is the Maximum exists.

**Result and Insight**

Key findings from the pricing simulation:

- Dynamic pricing increased profit by 91% compared to static pricing
- Peak demand occurs between 17:00 – 21:00
- Dynamic prices increase during high demand periods
- Demand elasticity ensures price increases do not significantly reduce usage

Business implication:

Dynamic pricing allows EV charging operators to align pricing with demand patterns, improving profitability while maintaining user retention**.**